FIT3080 Applied Class on Problem Solving as Search:
Graphsearch – Heuristic Search

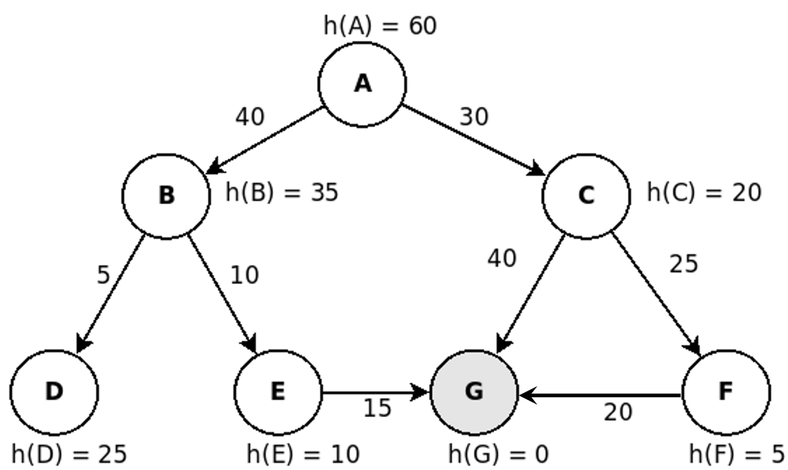
Exercise 1: Algorithm A/A*, Admissibility

(a) Is Algorithm A the same Algorithm as A*?

(b) Show that if $h(n)$ overestimates $h^*(n)$, Algorithm A may return a non-optimal solution. You can use an example.

Exercise 2: Informed and Uninformed Search Strategies

Consider the (full) state space below, where the goal node is **G**. For each of the following search strategies, show the FRONTIER in the order nodes are examined (IDDFS) or in order of merit (Greedy BFS, A and A*), which nodes remain in memory after the search ends, and what is the solution and its cost. For IDDFS also indicate the order in which nodes are created, and for Greedy BFS, Algorithm A and A*, also show EXPLORED.



(a) Iterative deepening depth-first search

(b) Greedy best-first search

(c) Algorithm A (as Graphsearch or Treesearch)

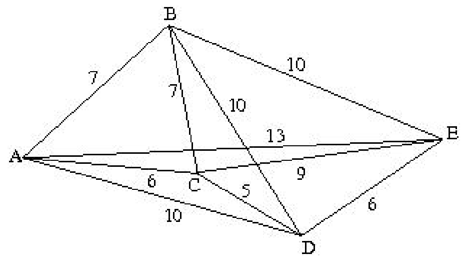
(d) What modifications are needed to turn Algorithm A into A*?

(e) After the modification, which nodes are expanded by A*?

Exercise 3: Informed Search Algorithms – Greedy-best-first-search/A/A*

Consider the Traveling Salesman Problem:

A salesman must visit each of n cities. There is a road between each pair of cities. Starting at city A, find the route of minimal distance that visits each of the cities only once and returns to city A.



(a) Propose two (non-zero) h functions for this problem. Is either of these h functions a lower bound of h^*?

(b) Apply algorithm A with these h functions to the 5 city problem above.

(c) How would your solution change if you used Greedy best first search instead of algorithm A?

Exercise 4: Algorithm A*

Consider a modified version of the 8-puzzle, where moving the blank tile to the center square has a cost of 4 (the rest of the moves cost 1). The initial configuration is:

2	8	3
1	6	4
7		5

Where the goal 8-puzzle is the following:

1	2	3
8		4
7	6	5

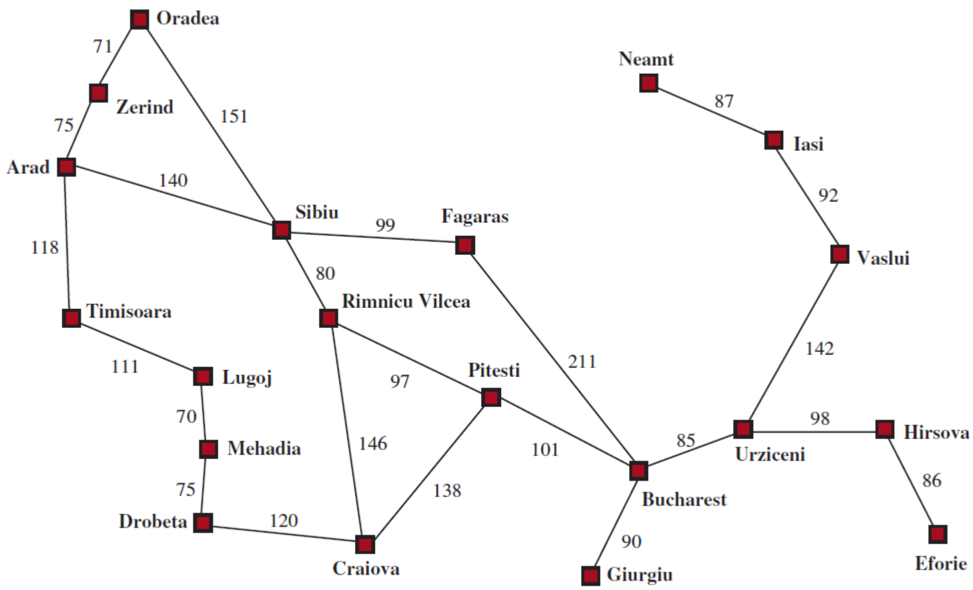
(a) Apply algorithm A* to find the shortest path between the initial and the final configuration. Use Manhattan Distance as your heuristic function.

In the Search Tree produced by the algorithm, indicate clearly the order of expansion of each node (i.e., when is a node **expanded**, not generated), and the values of g, h and f. Label the moves clearly.

(b) Describe what happens upon reaching the goal state. If you think that A* requires any modifications to improve its efficiency for this problem, describe and justify the modifications you envisage.

Exercise 5: Algorithm A* (Optional, Python Coding)

Once again, consider the Romania problem of finding a path from Arad to Bucharest.



Using the edge costs and the heuristic function from `lab3_astar_template.py` on Moodle, implement the A* algorithm to find a path from Arad to Bucharest.